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REVIEW ARTICLE (META-ANALYSIS)

Effect of Virtual Reality-Based Rehabilitation on Mental Health and Quality of Life of Stroke Patients: A Systematic Review and Meta-analysis of Randomized Controlled Trials

Check for updates

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Abstract

Objectives: To conduct a meta-analysis to investigate the effect of virtual reality (VR)-based rehabilitation on the mental health and quality of life of stroke patients.

Data Sources: The search strategy was conducted in 5 databases (PubMed, Scopus, Web of Science, Embase, and Cochrane Library databases) from inception to December 2023.

Study Selection: Randomized controlled trials (RCTs) comparing the effectiveness of standard rehabilitation and VR-based rehabilitation for stroke patients.

Data Extraction: Data from the included articles were extracted independently by 2 authors, with any disagreements resolved through consultation with a third author. The extracted data included the first author's name, country/region, publication year, sample size, mean/median age of participants, sex distribution (the proportion of males), VR type, duration of rehabilitation, comparison, intervention, and assessment of outcome.

Data Synthesis: A total of 29 studies involving 1561 stroke patients were included. The results showed that compared with standard rehabilitation, VR-based rehabilitation remarkably reduced anxiety symptoms [SMD=−0.97 (95% CI [−1.84, −0.09], P<.0001)], depression symptoms [SMD=−0.94 (95% CI [−1.46, −0.42], P<.001)], and improved quality of life [SMD=0.94 (95% CI [0.42, 1.45], P<.001)] of stroke patients. Subgroup analysis showed that immersive VR was particularly effective in reducing anxiety and depression symptoms compared to nonimmersive VR. The longer the duration of VR intervention, exceeding 6 weeks, the more significant the effect of improving anxiety and depression symptoms. Meanwhile, VR-based rehabilitation significantly improved the psychological state and quality of life of European patients.

Conclusions: VR-based rehabilitation significantly reduces anxiety and depression symptoms and enhances the quality of life in stroke patients compared to standard rehabilitation. The most notable improvements were observed with immersive VR-based rehabilitation programs over 6 weeks in duration, particularly among European patients.

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Stroke caused by cerebral infarction or hemorrhage is the main cause of permanent disability in adults and the second leading cause of death worldwide.[1] Over 25 million people worldwide are diagnosed with stroke, resulting in 6.5 million deaths each year.[2]

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Stroke patients often face physical disabilities along with psychological issues like anxiety and depression, which further hinder rehabilitation and affect quality of life.[3] Anxiety symptoms occur in about 24.2% of patients within the first year post-stroke,[4] while depression affects approximately 33%.[5] For stroke patients, rehabilitation therapy is a comprehensive intervention that significantly improves prognosis and quality of life. In addition, unlike psychological therapies that directly target anxiety and depression through cognitive-behavioral and interpersonal therapy,[6,7]

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rehabilitation therapy employs a multidisciplinary approach, including physical, occupational, speech therapy, psychological support, and so on. This holistic method aids in physical recovery and indirectly alleviates anxiety and depression by enhancing self-confidence and social support.[8] However, participation rates in traditional rehabilitation are low because of insufficient personalization and interactivity.[9]

Recently, virtual reality (VR) has been increasingly utilized in neurological rehabilitation for stroke patients.[10] VR is a computer-generated technology that simulates real-life scenarios to provide a novel rehabilitation environment that aids in restoring motor function, enhancing neurological regeneration, and encouraging active participation in rehabilitation.[11] VR technology showed effectiveness in improving mental health[12] and offers an innovative approach to stroke rehabilitation by simulating various actions for personalized and engaging experiences.[13] However, there is no consensus on the effect of VR-based rehabilitation on the mental health of stroke patients. A previous systematic review and meta-analysis study showed that using VR technology in stroke rehabilitation can improve patients' depression.[14] Another systematic review and meta-analysis study suggested that VR-based rehabilitation has no significant effect on overall depression.[15] In addition, previous meta-analyses have not explored the effect of VR type, rehabilitation intervention period, region, and patients' age on the rehabilitation outcomes of stroke patients.

Therefore, this study aims to reveal the effects of VR-based rehabilitation on the mental health and quality of life of stroke patients through a systematic review and meta-analysis, to provide new ideas and methods for stroke rehabilitation treatment and further expand the prospects for the application of VR technology in the field of rehabilitation.

Materials and methods

Protocol and search strategy

This systematic review and meta-analysis followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines,[16] with a detailed checklist in supplemental table S1. The study protocol has been registered with PROSPERO (registration number: CRD42024462333).

Relevant studies published until December 13, 2023, were identified by searching PubMed, Scopus, Web of Science, Embase, and Cochrane Library databases using predefined search strategies related to VR and stroke, along with Boolean operators. The detailed search strategies are provided in supplemental table S2. In addition, a manual search of reference lists in the included studies and the relevant reviews was performed to ensure the comprehensiveness of the literature search.

List of abbreviations:

CI	**confidence interval**
HADS	**Hospital Anxiety and Depression Scale**
PRISMA	**Preferred Reporting Items for Systematic Reviews and Meta-Analyses**
RCTs	**randomized controlled trials**
SMD	**standard mean difference**
SS-QOL	**Stroke-Specific Quality of Life Scale**
VR	**virtual reality**

Inclusion and exclusion criteria

Inclusion criteria are (1) adults with stroke (age≥18 years old); (2) participants undergoing VR-based rehabilitation. (VR technology was used in rehabilitation as an intervention through head-mounted displays and augmented reality glasses that created immersive, interactive environments simulating real-world scenarios. Alternatively, it might involve computer screens displaying virtual environments where patients perform rehabilitation exercises using game controllers and motion-sensing devices like Nintendo Wii and Microsoft Kinect); (3) study outcomes included levels of anxiety symptoms, levels of depression symptoms, or quality of life; (4) studies were randomized controlled trials (RCTs); and (5) published in English.

Exclusion criteria (1) pregnant women; (2) animal studies; (3) participants with other serious mental illnesses; and (4) conference abstracts, reviews, case reports, notes, letters, etc.

Data extraction process

Data for the included articles were extracted independently by 2 authors. In case of disagreement, a third author was consulted for resolution. The mentioned data were extracted from the included studies: the first author, country/region, publication year, sample, mean/median age of participants, sex (the proportion of males), VR type, duration of rehabilitation, comparison, intervention, and assessment of outcome.

Outcome measures

The scales used to assess anxiety, depression symptoms, and quality of life varied across studies. Anxiety and depression symptom scores were assessed using validated standardized questionnaires such as the Hospital Anxiety and Depression Scale (HADS),[17] while the quality of life was assessed using the Stroke-Specific Quality of Life Scale (SS-QOL)[18] or a comparable validated scale. The differences in anxiety, depression, and quality of life scores between the start and end of the rehabilitation were extracted and analyzed using standardized mean difference (SMD).[19]

Assessment for quality of studies

The quality of each included study was independently assessed by 2 reviewers using the Risk of Bias for Randomized Trials tool (RoB 2.0).[20] Disagreements were resolved by consensus or third-party consultation. The evaluation focused on 5 aspects: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Details of each aspect are shown in supplemental table S3. Each domain is rated as low, medium (some concerns), or high risk of bias. The overall risk of bias was then determined based on the combined assessments of these domains. Studies with all domains assessed as 'low risk' or no more than 1 domain with "some concerns" were considered high quality. Studies with up to 3 domains rated as "some concerns" were of moderate quality. Studies with more than 3 domains rated as "some concerns" or any "high risk" were considered as low quality.

Statistical analyses

Statistical analyses were performed using the Stata software (version 12.0 SE) and Review Manager (Version 5.3). The effect sizes were defined as the weighted SMD and 95% confidence interval (CI). Means and SD were used to calculate overall effect sizes.

The I^2 index was used to assess heterogeneity. I^2 value less than 25 % represented low heterogeneity, from 25% to 50% represented medium heterogeneity, and greater than 50% represented high heterogeneity.[21] A random effects model was applied when heterogeneity was high ($P<.10$ or $I^2>50\%$). Subgroup analyses were performed to explore the source of heterogeneity and to determine whether certain specific characteristics influenced the overall effect size. In this study, the data were grouped by region, sample size, sex, age, type of VR, and duration of rehabilitation to determine whether these factors affect the effectiveness of VR-based rehabilitation. Sensitivity analysis was used to assess the effect of each study on overall significance. Funnel plots and Egger's linear regression test were performed to evaluate the publication bias. The P value <.05 was considered statistically significant.

Results

Literature search and selection

The preliminary systematic search identified 14,401 studies from online databases. An additional 41 studies were identified through manual searches. After manually removing duplicates, 8232 articles remained. Title and abstract screening excluded 8080 articles because of animal experiments; irrelevance; non-English publications; conference abstracts, reviews, letters, or comments. The full-text evaluation further excluded 152 articles for nonadult participants (n=4), irrelevant outcomes (n=80), non-English studies (n=17), and non-RCTs (n=26). Finally, 25 articles were included in this systematic review and meta-analysis. The detailed literature search process is shown in figure 1.

Characteristics of the included studies

This meta-analysis included 25 studies involving 1150 stroke patients, published between 2015 and 2023. Sample sizes ranged from 16 to 145, with participants' mean/median ages from 40.1 to 72.0 years, and male proportions varying from 29.2% to 81.4%. The mean/median follow-up time ranged from 3 to 24 weeks. Among the studies, 4 assessed anxiety symptoms,[22-25] 13 examined depression symptoms,[22-24,26-35] and 15 evaluated quality of life.[22,25,32,35-46] Detailed study information was presented in supplemental table S4.

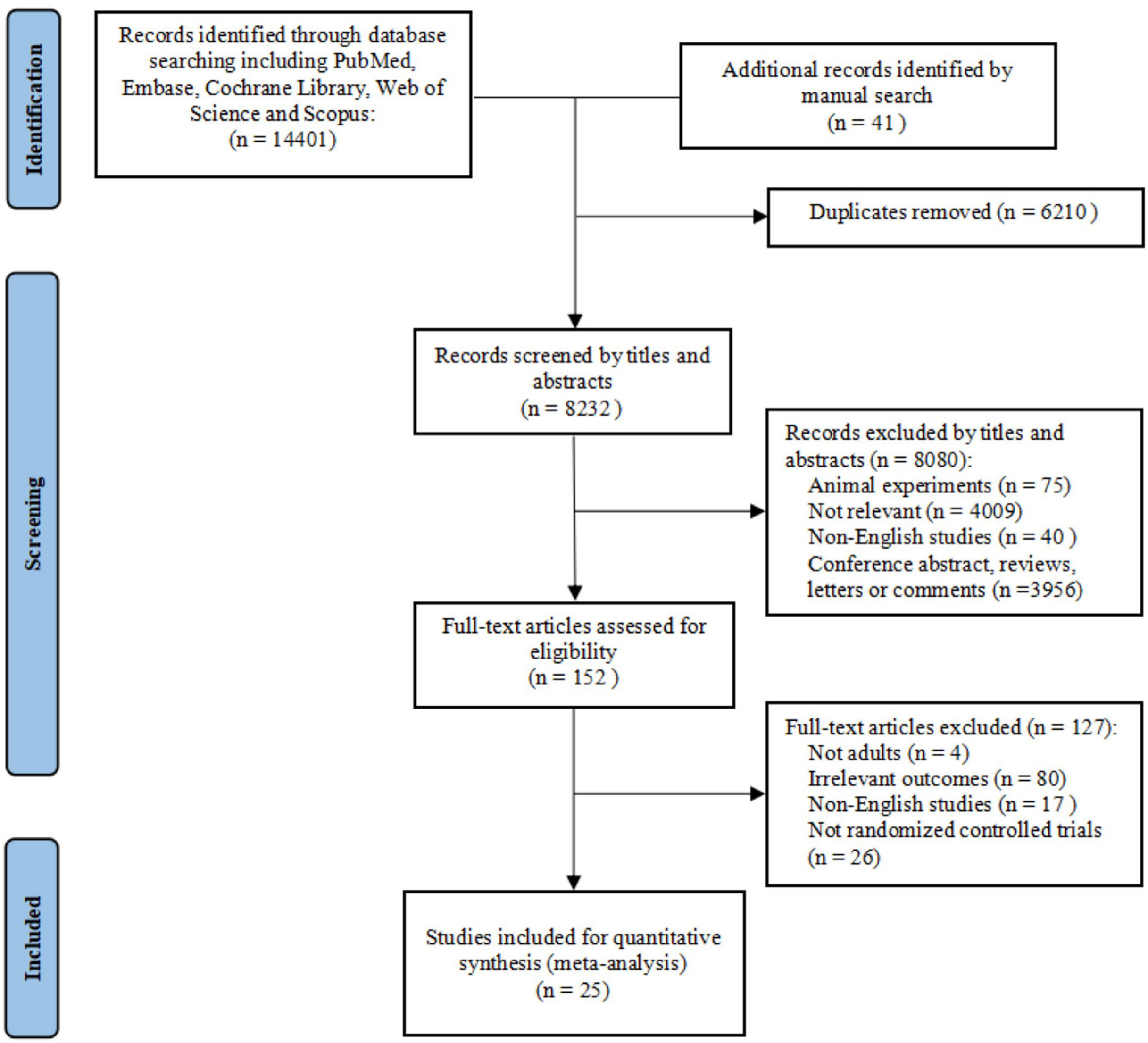

Fig 1 Flowing diagram of included studies selection process.

Risk of bias assessment

The risk of bias, assessed using the RoB 2.0 tool, indicated 13 with a low risk of bias, 6 with some concerns, and 6 with a high risk of bias (fig 2).

Effect of VR-based rehabilitation on anxiety symptoms

As shown in figure 3, 4 studies involved 300 stroke patients focused on anxiety symptoms. Among these studies, 2 were of

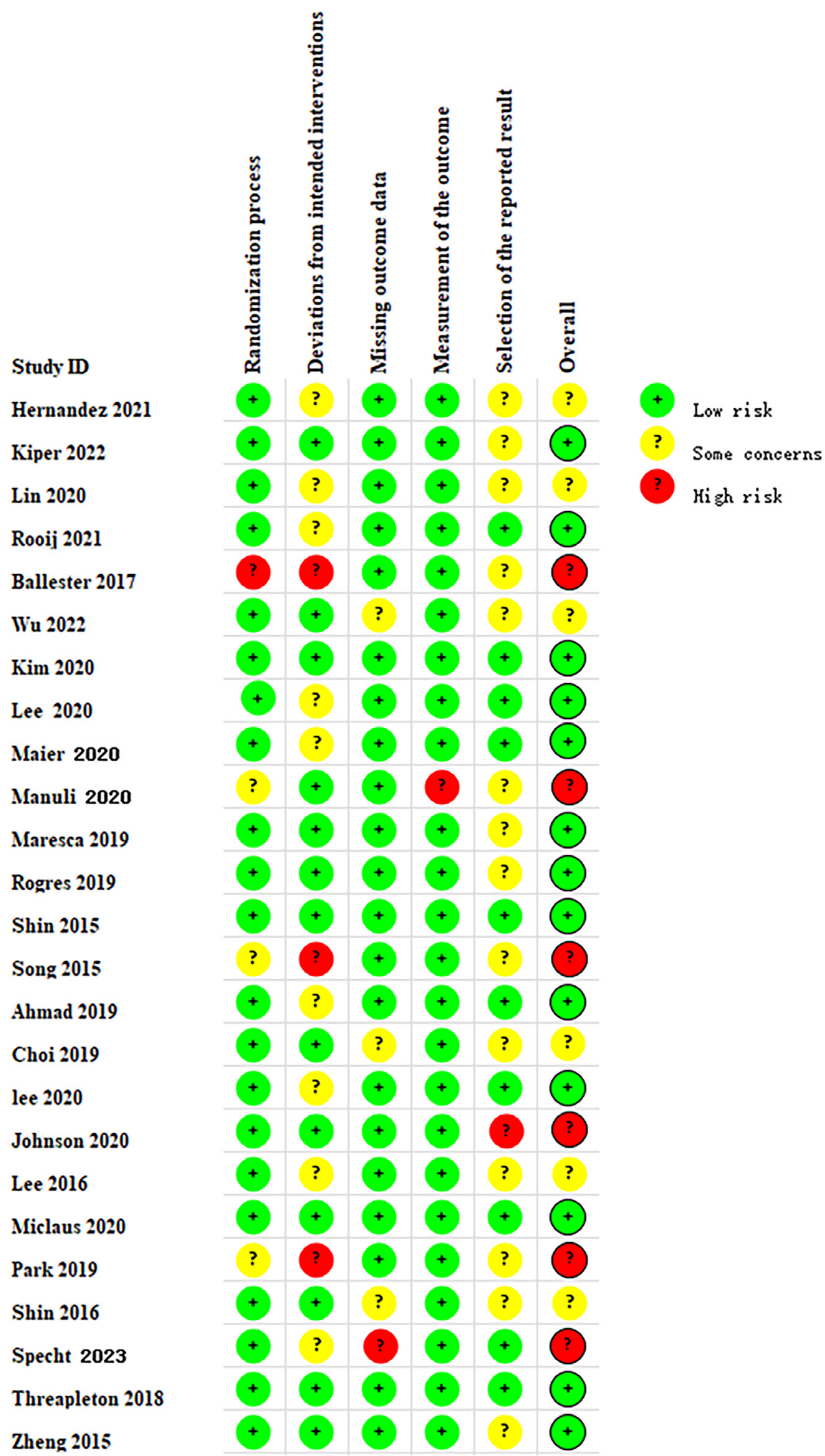

Fig 2 The risk of bias of included randomized controlled trials.

	Experimental			Control				Std. Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Hernandez 2021	-4.9	0.7	23	-3.3	0.4	20	22.3%	-2.71 [-3.55, -1.86]
Kiper 2022	-2.25	1.8	30	-2.11	1.85	30	25.8%	-0.08 [-0.58, 0.43]
Lin 2020	-1.8	2.2	38	-0.2	5	107	26.8%	-0.36 [-0.73, 0.01]
Rooij 2021	0.02	0.24	28	0.27	0.26	24	25.1%	-0.99 [-1.57, -0.41]
Total (95% CI)			119			181	100.0%	-0.97 [-1.84, -0.09]

Heterogeneity: Tau² = 0.70; Chi² = 31.12, df = 3 (P < 0.00001); I² = 90%
Test for overall effect: Z = 2.17 (P = 0.03)

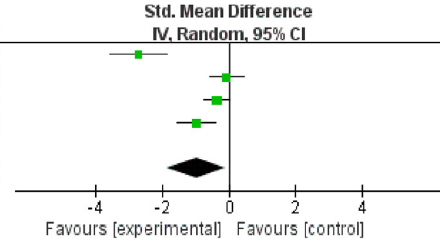

Fig 3 Forest plot indicating the effect of virtual reality-based rehabilitation on anxiety symptoms in stroke patients.

high quality and 2 were of moderate quality. The present meta-analysis, using a random effects model, showed that VR-based rehabilitation significantly improved anxiety symptoms in stroke patients compared to standard rehabilitation [SMD=−0.97 (95% CI [−1.84, −0.09], P<.0001), I^2=90%].

Effect of VR-based rehabilitation on depression symptoms

Thirteen studies involving 603 stroke patients reported depression symptoms as the main outcome (fig 4). Among these, 6 were of high quality, 2 were of moderate quality, and 5 were of low quality. The random effects model analysis indicated that VR-based rehabilitation statistically reduced the level of depression symptoms in stroke patients compared to standard rehabilitation [SMD=−0.94 (95% CI [−1.46, −0.42], P<.001), I^2=87%].

Effect of VR-based rehabilitation on quality of life

Fifteen studies involving 658 stroke patients assessed quality of life. Of the 15 studies, 9 were of high quality, 4 of moderate quality, and 2 of low quality. As shown in figure 5, the meta-analysis using a random effects model showed that VR-based rehabilitation statistically improved the quality of life in stroke patients compared to standard rehabilitation [SMD=0.94 (95% CI [0.42, 1.45], P<.001), I^2=89%].

Subgroup analysis

Subgroup analyses were performed to explore sources of heterogeneity and determine whether certain specific characteristics influenced the overall effect size. The data were grouped by region, sample size, sex, age, type of VR, and duration of rehabilitation (table 1). Subgroup analyses indicated that VR-based rehabilitation significantly improved mental health and quality of life in stroke patients regardless of these factors. This consistency supports the robustness of the finding that VR-based rehabilitation is more effective than standard rehabilitation in improving the mental health and quality of life in stroke patients. In addition, the subgroup analyses showed that VR-based rehabilitation had the most significant effect on improving anxiety, depression, and quality of life in Europe. Furthermore, immersive VR showed a more pronounced improvement in patients' levels of anxiety and depression compared to nonimmersive VR. In addition, VR-based rehabilitation interventions lasting 6 weeks or longer were notably effective in reducing depression levels in stroke patients.

Sensitivity analysis

Leave-one-out sensitivity analyses were conducted to assess the effect of individual trials on overall results (fig 6). The sensitivity analysis indicated that no single study significantly influenced the outcomes for anxiety symptoms, depression symptoms, or quality of life in stroke patients. These analyses confirm that VR technology significantly reduces anxiety and depression symptoms while improving the quality of life in stroke patients.

Publication bias and heterogeneity

Publication bias was analyzed through funnel plots and Egger's test (fig 7). Visual inspection of the funnel plots for anxiety and depression symptoms showed symmetry, with Egger's test P

	Experimental			Control				Std. Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Ballester 2017	-1.64	1.57	17	-1.18	2.23	18	8.0%	-0.23 [-0.90, 0.43]
J.-J. Wu 2022	-4.82	2.89	22	-0.96	1.9	22	8.0%	-1.55 [-2.23, -0.87]
Kim 2020	-3.38	1.46	12	-4.33	1.15	12	7.5%	0.70 [-0.13, 1.53]
Kiper 2022	-6.6	1.91	30	-3.17	1.91	30	8.2%	-1.77 [-2.38, -1.17]
Lee 2020	-1.64	1.57	11	-1.18	2.23	11	7.4%	-0.23 [-1.07, 0.61]
Lin 2020	-2.8	1.9	38	-0.3	4.7	107	8.8%	-0.60 [-0.97, -0.22]
Maier 2020	-2.45	4.32	19	1.1	2.18	19	8.0%	-1.02 [-1.70, -0.34]
Manuli 2020	-7.4	4.2	30	-1.3	3.9	30	8.3%	-1.49 [-2.06, -0.91]
Maresca 2019	-6.5	0.68	15	-2.3	0.68	15	4.4%	-6.01 [-7.79, -4.23]
Rogers 2019	-8.1	4.4	10	-3.4	6.8	11	7.2%	-0.78 [-1.67, 0.12]
Rooij 2021	-0.15	0.3	28	-0.02	0.32	24	8.4%	-0.41 [-0.97, 0.14]
Shin 2015	-0.5	1.8	16	-1.5	1.4	16	7.9%	0.60 [-0.11, 1.32]
Song 2015	-7.1	3.3	20	-2.1	3	20	7.9%	-1.55 [-2.27, -0.84]
Total (95% CI)			268			335	100.0%	-0.94 [-1.46, -0.42]

Heterogeneity: Tau² = 0.76; Chi² = 92.69, df = 12 (P < 0.00001); I² = 87%
Test for overall effect: Z = 3.55 (P = 0.0004)

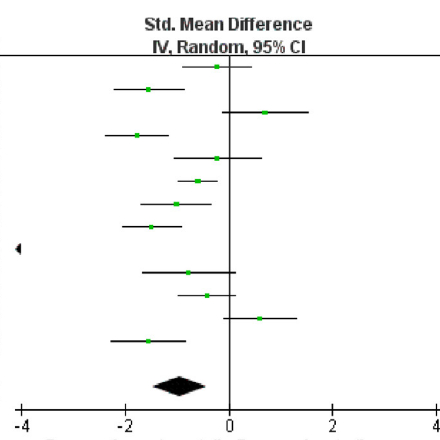

Fig 4 Forest plot indicating the effect of virtual reality-based rehabilitation on depression symptoms in stroke patients.

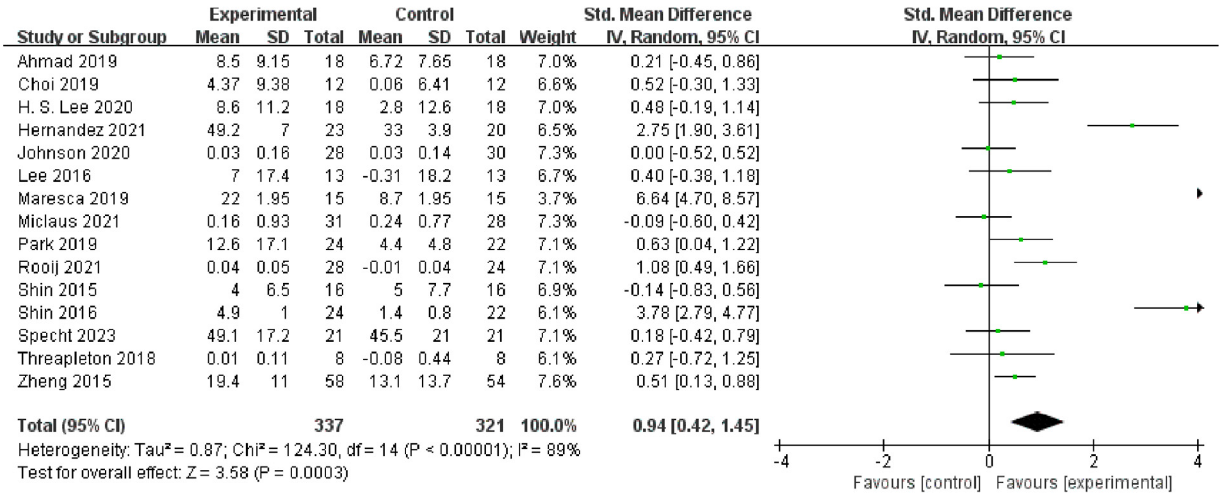

Fig 5 Forest plot indicating the effect of virtual reality-based rehabilitation on quality of life in stroke patients.

values of .154 and .337, indicating no publication bias. However, the funnel plot for quality of life showed significant asymmetry, with a *P* value of .012, suggesting potential publication bias. Therefore, the trim and fill method was applied (fig 8). After adding dummy studies, the result remained statistically significant (recalculated SMD=2.547, 95% CI [1.53, 4.25]), indicating that the result was not affected by publication bias.

The meta-analysis results indicated high heterogeneity (anxiety symptoms: I^2=90%, depression symptoms: I^2=87%, quality of life: I^2=89%). Subgroup analysis was performed to explore potential sources of heterogeneity, revealing that region, VR type, sex, and the intervention period may be sources of heterogeneity.

Discussion

This systematic review and meta-analysis demonstrated that VR-based rehabilitation significantly improved mental health and quality of life in stroke patients compared to standard rehabilitation. Notably, immersive VR had a more pronounced effect on reducing anxiety and depression levels than nonimmersive VR. Furthermore, interventions lasting 6 weeks or more significantly enhanced depression levels in stroke patients.

Contrary to a previous systematic review that indicated no significant effect of VR on improving depression and quality of life in stroke patients,[15] the present study found that VR-based rehabilitation significantly improved both depression symptoms and quality of life compared to standard rehabilitation. This discrepancy may be because of the fact that the former review included only 5 articles published before 2021, which may have resulted in smaller effect sizes. In addition, meta-analysis by Lin et al. found that VR games reduced depression symptoms in stroke patients,[14] although their study focused exclusively on elderly stroke patients. The present study aligns with previous research,[14,47,48] indicating that VR-based rehabilitation could improve depression symptoms in stroke patients, while also examining its effects on anxiety symptoms and quality of life. Meanwhile, the present study deeply explored the reasons for the improvement in mental health and quality of life in stroke patients through VR-based rehabilitation and conducted subgroup analyses to explore the potential factors that might affect these outcomes. VR-based rehabilitation therapy has been shown to improve anxiety, depression symptoms, and quality of life in stroke patients, which could be attributed to the mentioned factors. First, VR can activate the hippocampus and amygdala, improve the interaction between brain structures and functional regions, jointly regulate emotional and cognitive functions, enhance neural remodeling, and thus improve symptoms of anxiety and depression.[49] In addition, simulating real-world activities in VR encourages neural plasticity, stimulating brain neuron activity and aiding in the recovery of neural function, which may enhance mental health and quality of life.[50]

Moreover, VR therapy is often vivid and engaging, boosting patient motivation and participation compared to traditional rehabilitation, which makes patients more willing to actively participate and easier to improve their anxiety, depression, and quality of life.[51] In addition, patients immersed in VR-based exercise games are likely to experience increased motivation, which activates the brain's reward circuits and leads to higher dopamine production, thereby alleviating anxiety and depression symptoms.[52,53]

In addition, the subgroup analysis showed that immersive VR-based rehabilitation improved anxiety and depression symptoms more significantly in stroke patients than non-immersive. The stronger sense of immersion in immersive VR allows patients to forget physical discomfort, enhancing focus on rehabilitation and increasing participation and enthusiasm.[54] This can promote their more active participation in rehabilitation training, thereby achieving better treatment outcomes. Immersive VR technology can be customized according to the needs of patients, providing more personalized and adaptive rehabilitation plans.[55] This can help patients more effectively address their rehabilitation issues, thereby improving their psychological condition and quality of life. Immersive VR environments can simulate various social situations, provide emotional support and connection, and make post-stroke patients feel no longer isolated.[56] By interacting with virtual characters or other patients, sharing experiences and emotions, patients can feel understanding and support, reducing the negative effects of anxiety and depression. It is also possible to improve anxiety and depression in post-stroke patients by regulating the autonomic nervous system and brain neural plasticity.[57]

Subgroup results also revealed that VR-based rehabilitation interventions lasting 6 weeks or longer significantly improved depression levels in stroke patients, which is consistent with long-term VR rehabilitation interventions that can provide sustained therapeutic effects.[58] Long-term VR-based rehabilitation plays a

Table 1 Results of subgroup analysis of included studies in the meta-analysis

Groups	n	Anxiety			n	Depression			n	Quality of Life		
		SMD (95% CI)	I^2 (%)	*P* Value		SMD (95% CI)	I^2 (%)	*P* Value		SMD (95% CI)	I^2 (%)	*P* Value
Region												
Europe	3	−0.86 (−1.21, −0.51)	93.0	.000	6	−1.13 (−1.40, −0.86)	90.2	.000	6	0.77 (0.48, 1.06)	93.5	.000
Asia	1	−0.36 (−0.73, 0.01)	-	-	6	−0.55 (−0.80, −0.30)	86.5	.000	8	0.58 (0.36, 0.80)	85.3	.000
Oceania	-	-	-	-	1	−0.81 (−1.17, −0.18)	-	-	1	0.0 (−0.52, 0.52)	-	-
Sample size												
<50	1	−2.76 (−3.60, −1.91)	-	-	9	−0.69 (−0.95, −0.43)	89.7	.000	11	0.79 (0.56, 1.02)	91.1	.000
≥50	3	−0.42 (−0.58, −0.15)	65.2	.057	4	−0.94 (−1.19, −0.69)	83.3	.000	4	0.37 (0.17, 0.60)	74.0	.009
Virtual reality type												
Immersive	2	−1.56 (−2.04, −1.08)	91.1	.001	5	−0.94 (−1.19, −0.69)	78.5	.002	2	0.31 (−0.17, 0.80)	0.0	.501
Nonimmersive	2	−0.26 (−0.56, −0.04)	0.0	.377	8	−0.69 (−0.95, −0.42)	90.9	.000	13	0.62 (0.45, 0.80)	90.7	.000
Age												
<65	2	−1.56 (−2.04, −1.08)	91.1	.001	9	−0.92 (−1.16, −0.69)	88.6	.000	9	0.84 (0.61, 1.08)	93.3	.000
≥65	2	−0.26 (−0.56, −0.04)	0.0	.377	4	−0.67 (−0.95, −0.39)	88.2	.000	5	0.42 (0.16, 0.68)	0.0	.925
Not-reported	-	-	-	-	-	-	-	-	1	0.0 (−0.52, 0.52)	-	-
Sex (Male%)												
<50	-	-	-	-	6	−0.88 (−1.19, −0.58)	91.3	.000	4	0.36 (−0.02, 0.75)	93.5	.000
≥50	4	−0.63 (−0.88, −0.37)	90.8	.000	7	−0.79 (−1.01, −0.57)	85.0	.000	10	0.73 (0.53, 0.92)	88.1	.000
Not-reported	-	-	-	-	-	-	-	-	1	0.0 (−0.52, 0.52)	-	-
Duration of rehabilitation (wks)												
<6	1	−2.76 (−3.60, −1.91)	-	-	6	−0.29 (−0.59, 0.02)	81.6	.000	6	0.69 (0.45, 0.92)	93.6	.000
≥6	3	−0.42 (−0.68, −0.15)	65.2	.057	7	−1.09 (−1.31, −0.87)	88.7	.000	6	0.55 (0.28, 0.83)	89.9	.000
Not-reported	-	-	-	-	-	-	-	-	3	0.31 (−0.13, 0.74)	0.0	.796

Abbreviations: CI, confidence interval; SMD, standardized mean difference.

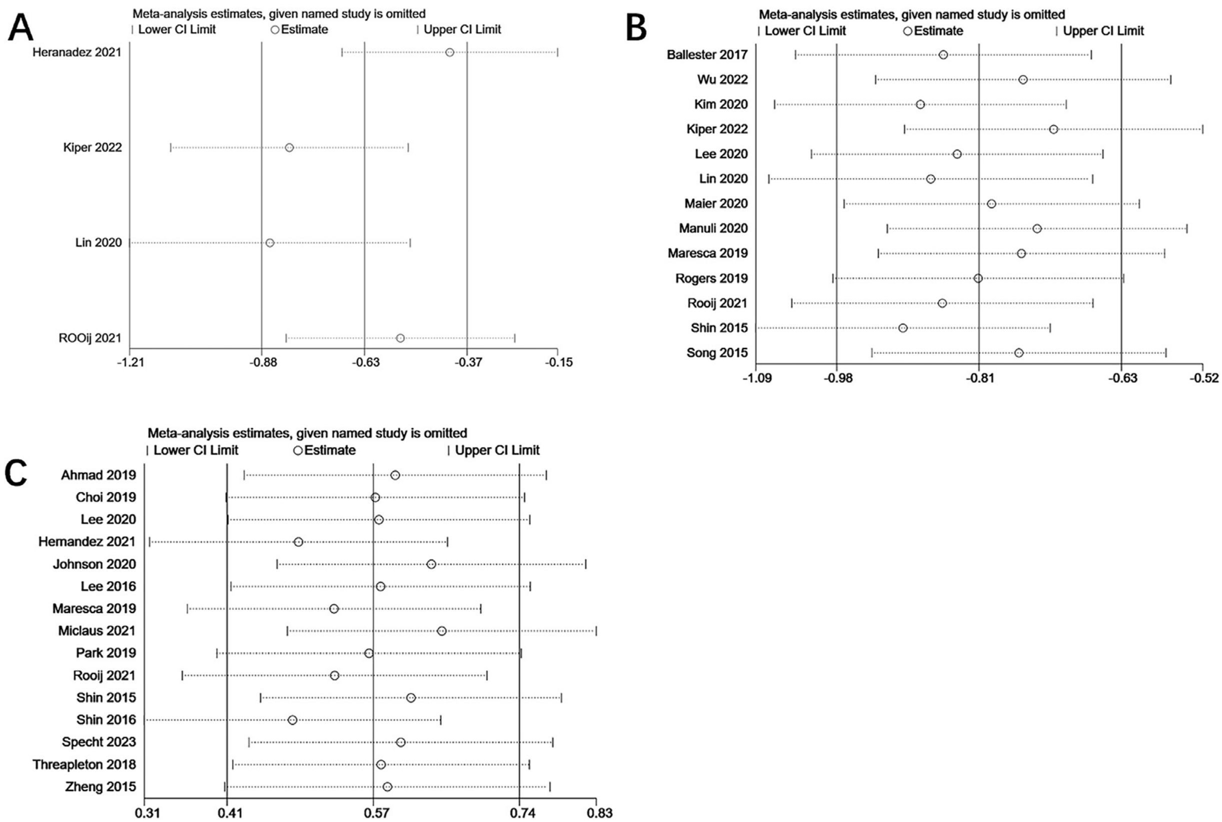

Fig 6 (**A**) Sensitivity analysis of anxiety symptoms. (**B**) Sensitivity analysis of depression symptoms. (**C**) Sensitivity analysis of quality of life.

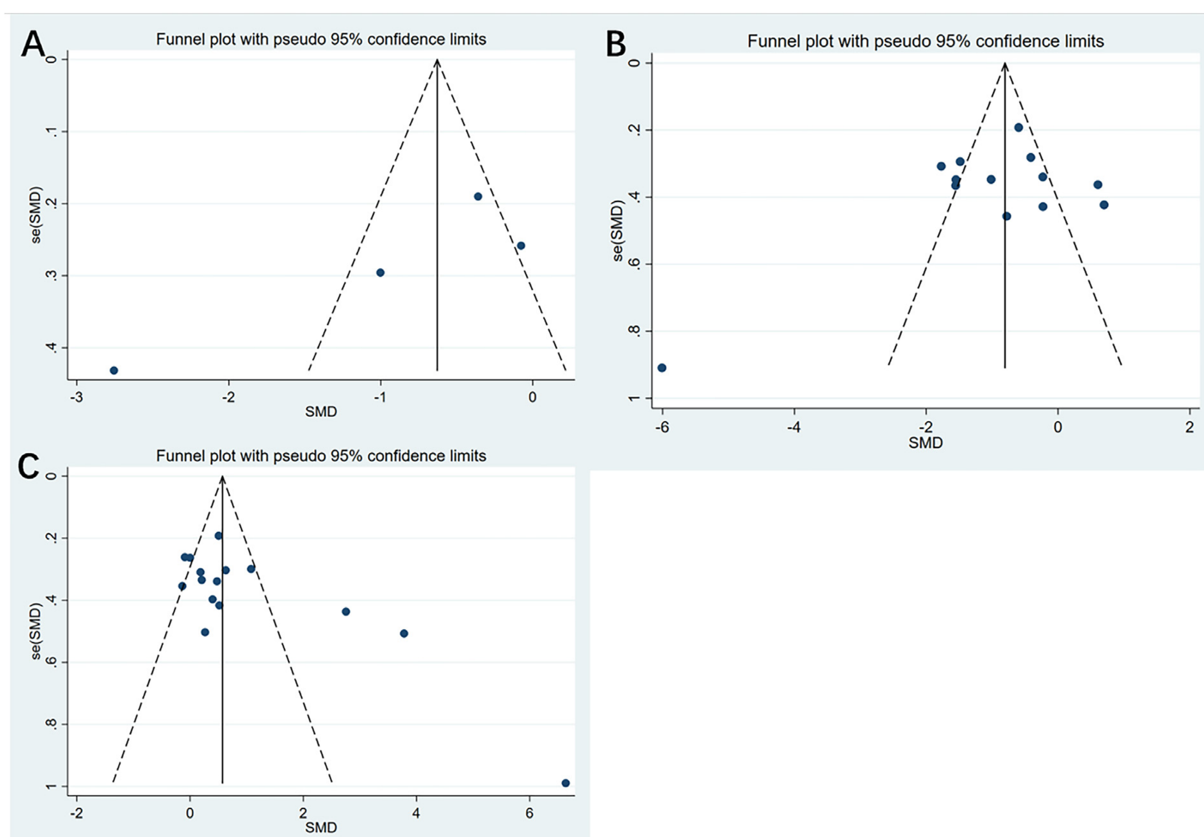

Fig 7 (A) Funnel plot of anxiety symptoms. (B) Funnel plot of depression symptoms. (C) Funnel plot of quality of life.

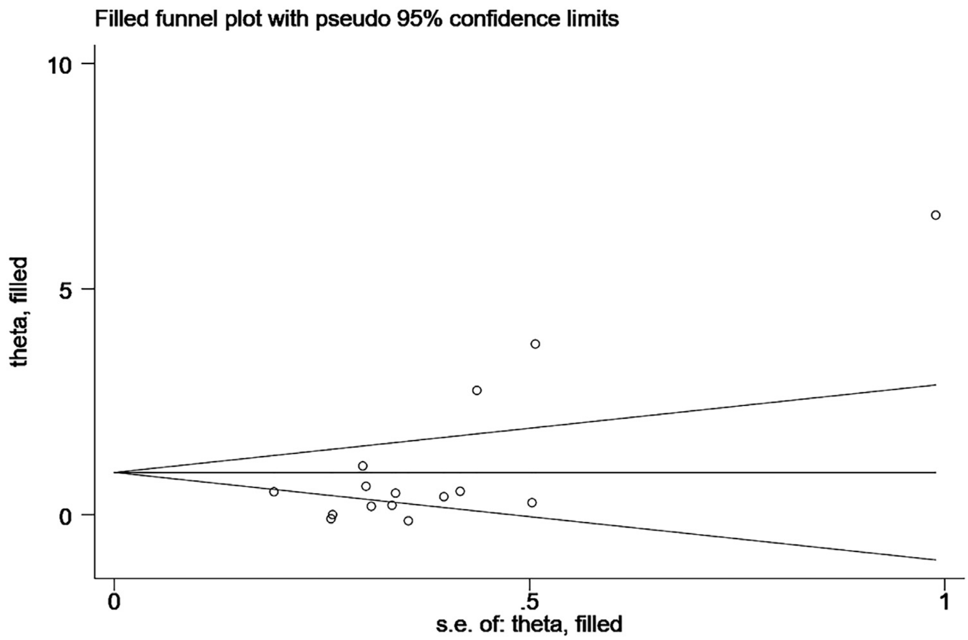

Fig 8 Trim and fill funnel plot of quality of life.

significant role in promoting brain neuroplasticity, a key factor in stroke recovery.[59] Neuroplasticity refers to the brain's ability to reorganize and restore impaired functions by forming new neural networks. Studies have shown that frequent and sustained VR-based rehabilitation training can significantly enhance neuroplasticity,[60,61] potentially leading to improved depressive symptoms in stroke patients. Moreover, long-term VR rehabilitation training can provide sustained psychological and emotional support, reducing patients' feelings of loneliness and anxiety.[62] VR technology can simulate social interactions and provide emotional experiences that help patients establish deeper emotional bonds, which is important for improving depressive symptoms.[63] The social support provided in virtual environments can supplement emotional support in real life, especially for patients who may feel isolated in their actual environments. Meanwhile, VR-based rehabilitation was found to have a more significant effect on anxiety, depression, and quality of life among European stroke patients. This is because of the fact that European countries are more supportive of the use of VR technology in health care,[64] which makes it easier for patients there to access VR therapy. VR technology is regarded as an advanced therapeutic tool that is also more likely to be recognized by health care providers and individuals in the European cultural context.[65] In addition, most of the studies conducted in Europe that were included in this study had rehabilitation intervention periods of 6 weeks or longer. This longer duration of VR-based rehabilitation may explain the greater effectiveness observed in European patients. Long-term VR-based rehabilitation can help to promote brain neuroplasticity and provide continuous psychological and emotional support to stroke patients, leading to improvements in mental health and quality of life.[59,62] Further RCTs with large sample sizes are needed in the future to explore the long-term benefits of VR-based rehabilitation for stroke patients.

Study limitations

Although this study found that VR-based rehabilitation significantly improved the mental health and quality of life of stroke patients compared with standard rehabilitation, there were several potential limitations should be noted. First, because of the limited number of included studies, the results of our meta-analysis may not be fully generalizable. Future large-scale studies with large sample sizes are needed to confirm our findings and enhance their robustness. Second, the type and severity of stroke could potentially affect the study results. Future research should explore the effects of VR-based rehabilitation on stroke patients with different types and severity of stroke. Third, differences in the devices or technologies used for VR interventions, as well as geographic variations, and disparities in health care system/reimbursement models may also affect the outcomes. Finally, the meta-analysis only includes articles published in English, which may not fully reflect global research on this topic. Despite these limitations, the study's large sample size and multifactor subgroup analyses and sensitivity analyses were performed, both of which showed robust meta-analysis results.

Conclusion

The present meta-analysis reveals that VR-based rehabilitation significantly reduces anxiety and depression symptoms and improves the quality of life of stroke patients compared with standard rehabilitation. The most notable improvements were observed with immersive VR-based rehabilitation programs over 6 weeks in European patients. In conclusion, VR-based rehabilitation emerges as a valuable and effective approach for stroke patients, especially in improving mental health and quality of life.

Authorship Contributions

Saikun Wang and Hongli Meng contributed to Conceptualization, Investigation, Methodology, Data curation, and Writing—original draft. Yong Zhang contributed to Conceptualization, Investigation, Data curation. Jing Mao contributed to Methodology. Changyue Zhang and Chunting Qian contributed to Investigation.

Yueping Ma contributed to Software. Lirong Guo contributed to Conceptualization, Supervision, and Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

Keywords

Mental health; Quality of life; Rehabilitation; Stroke; Virtual reality

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